burnout project

February 24, 2026

[1]:
```
pip install pandas numpy matplotlib seaborn scikit-learn statsmodels prophet
```

```
Requirement already satisfied: pandas in c:\users\rpere\anaconda3\lib\site-
packages (1.5.3)Note: you may need to restart the kernel to use updated
packages.

Requirement already satisfied: numpy in c:\users\rpere\anaconda3\lib\site-
packages (1.24.3)
Requirement already satisfied: matplotlib in c:\users\rpere\anaconda3\lib\site-
packages (3.7.1)
Requirement already satisfied: seaborn in c:\users\rpere\anaconda3\lib\site-
packages (0.12.2)
Requirement already satisfied: scikit-learn in
c:\users\rpere\anaconda3\lib\site-packages (1.3.2)
Requirement already satisfied: statsmodels in c:\users\rpere\anaconda3\lib\site-
packages (0.14.0)
Requirement already satisfied: prophet in c:\users\rpere\anaconda3\lib\site-
packages (1.3.0)
Requirement already satisfied: python-dateutil>=2.8.1 in
c:\users\rpere\anaconda3\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
c:\users\rpere\anaconda3\lib\site-packages (from pandas) (2022.7)
Requirement already satisfied: contourpy>=1.0.1 in
c:\users\rpere\anaconda3\lib\site-packages (from matplotlib) (1.0.5)
Requirement already satisfied: cycler>=0.10 in
c:\users\rpere\anaconda3\lib\site-packages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in
c:\users\rpere\anaconda3\lib\site-packages (from matplotlib) (4.25.0)
Requirement already satisfied: kiwisolver>=1.0.1 in
c:\users\rpere\anaconda3\lib\site-packages (from matplotlib) (1.4.4)
Requirement already satisfied: packaging>=20.0 in
c:\users\rpere\anaconda3\lib\site-packages (from matplotlib) (23.0)
Requirement already satisfied: pillow>=6.2.0 in
c:\users\rpere\anaconda3\lib\site-packages (from matplotlib) (9.4.0)
Requirement already satisfied: pyparsing>=2.3.1 in
c:\users\rpere\anaconda3\lib\site-packages (from matplotlib) (3.0.9)
Requirement already satisfied: scipy>=1.5.0 in
c:\users\rpere\anaconda3\lib\site-packages (from scikit-learn) (1.10.1)
```

```
Requirement already satisfied: joblib>=1.1.1 in
c:\users\rpere\anaconda3\lib\site-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=2.0.0 in
c:\users\rpere\anaconda3\lib\site-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: patsy>=0.5.2 in
c:\users\rpere\anaconda3\lib\site-packages (from statsmodels) (0.5.3)
Requirement already satisfied: cmdstanpy>=1.0.4 in
c:\users\rpere\anaconda3\lib\site-packages (from prophet) (1.3.0)
Requirement already satisfied: holidays<1,>=0.25 in
c:\users\rpere\anaconda3\lib\site-packages (from prophet) (0.91)
Requirement already satisfied: tqdm>=4.36.1 in
c:\users\rpere\anaconda3\lib\site-packages (from prophet) (4.65.0)
Requirement already satisfied: importlib_resources in
c:\users\rpere\anaconda3\lib\site-packages (from prophet) (6.5.2)
Requirement already satisfied: stanio<2.0.0,>=0.4.0 in
c:\users\rpere\anaconda3\lib\site-packages (from cmdstanpy>=1.0.4->prophet)
(0.5.1)
Requirement already satisfied: six in c:\users\rpere\anaconda3\lib\site-packages
(from patsy>=0.5.2->statsmodels) (1.16.0)
Requirement already satisfied: colorama in c:\users\rpere\anaconda3\lib\site-
packages (from tqdm>=4.36.1->prophet) (0.4.6)
```

[3]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# display settings
pd.set_option('display.max_columns', None)
pd.set_option('display.max_rows', 10)
```

[8]:
```python
df = pd.read_csv("burnot_cleann.csv")
```

[9]:
```python
df.head()
```

[9]:
```
                Employee ID Date of Joining  Gender Company Type  \
0  fffe32003000360033003200      30-09-2008  Female      Service
1      fffe3700360033003500      30-11-2008    Male      Service
2  fffe32003400380032003900      03-11-2008    Male      Service
3  fffe31003900340031003600      24-07-2008  Female      Service
4      fffe3300350037003500      26-11-2008    Male      Product

  WFH Setup Available  Designation  Resource Allocation  Mental Fatigue Score  \
0                  No            2                    3                   3.8
1                 Yes            1                    2                   5.0
2                 Yes            1                    1                   2.6
3                  No            3                    7                   6.9
4                 Yes            2                    4                   3.6
```

```
   Burn Rate
0       0.16
1       0.36
2       0.20
3       0.52
4       0.29
```

[10]:
```
df.shape
```

[10]:
```
(18590, 9)
```

[11]:
```
df.columns
```

[11]:
```
Index(['Employee ID', 'Date of Joining', 'Gender', 'Company Type',
       'WFH Setup Available', 'Designation', 'Resource Allocation',
       'Mental Fatigue Score', 'Burn Rate'],
      dtype='object')
```

[12]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 18590 entries, 0 to 18589
Data columns (total 9 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Employee ID           18590 non-null  object
 1   Date of Joining       18590 non-null  object
 2   Gender                18590 non-null  object
 3   Company Type          18590 non-null  object
 4   WFH Setup Available   18590 non-null  object
 5   Designation           18590 non-null  int64
 6   Resource Allocation   18590 non-null  int64
 7   Mental Fatigue Score  18590 non-null  float64
 8   Burn Rate             18590 non-null  float64
dtypes: float64(2), int64(2), object(5)
memory usage: 1.3+ MB
```

[13]:
```
df.isnull().sum()
```

[13]:
```
Employee ID             0
Date of Joining         0
Gender                  0
Company Type            0
WFH Setup Available     0
Designation             0
Resource Allocation     0
Mental Fatigue Score    0
```

```
Burn Rate                0
dtype: int64
```

[14]:
```python
df.describe()
```

[14]:

	Designation	Resource Allocation	Mental Fatigue Score	Burn Rate
count	18590.000000	18590.000000	18590.000000	18590.000000
mean	2.179398	4.486552	5.732173	0.452444
std	1.133148	2.044848	1.920547	0.197848
min	0.000000	1.000000	0.000000	0.000000
25%	1.000000	3.000000	4.600000	0.320000
50%	2.000000	4.000000	5.900000	0.450000
75%	3.000000	6.000000	7.100000	0.590000
max	5.000000	10.000000	10.000000	1.000000

[15]:
```python
df.duplicated().sum()
```

[15]: 0

[16]:
```python
df.to_csv("Burnout_Phase2_Loaded.csv", index=False)
```

[17]:
```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 18590 entries, 0 to 18589
Data columns (total 9 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Employee ID           18590 non-null  object
 1   Date of Joining       18590 non-null  object
 2   Gender                18590 non-null  object
 3   Company Type          18590 non-null  object
 4   WFH Setup Available   18590 non-null  object
 5   Designation           18590 non-null  int64
 6   Resource Allocation   18590 non-null  int64
 7   Mental Fatigue Score  18590 non-null  float64
 8   Burn Rate             18590 non-null  float64
dtypes: float64(2), int64(2), object(5)
memory usage: 1.3+ MB
```

[18]:
```python
df['Date of Joining'] = pd.to_datetime(df['Date of Joining'])
```

```
C:\Users\rpere\AppData\Local\Temp\ipykernel_54020\3163426663.py:1: UserWarning:
Parsing dates in DD/MM/YYYY format when dayfirst=False (the default) was
specified. This may lead to inconsistently parsed dates! Specify a format to
ensure consistent parsing.
  df['Date of Joining'] = pd.to_datetime(df['Date of Joining'])
```

[19]:
```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 18590 entries, 0 to 18589
Data columns (total 9 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Employee ID           18590 non-null  object
 1   Date of Joining       18590 non-null  datetime64[ns]
 2   Gender                18590 non-null  object
 3   Company Type          18590 non-null  object
 4   WFH Setup Available   18590 non-null  object
 5   Designation           18590 non-null  int64
 6   Resource Allocation   18590 non-null  int64
 7   Mental Fatigue Score  18590 non-null  float64
 8   Burn Rate             18590 non-null  float64
dtypes: datetime64[ns](1), float64(2), int64(2), object(4)
memory usage: 1.3+ MB
```

[20]:
```
df['Date of Joining'] = pd.to_datetime(df['Date of Joining'], format='%d-%m-%Y')
```

[21]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 18590 entries, 0 to 18589
Data columns (total 9 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Employee ID           18590 non-null  object
 1   Date of Joining       18590 non-null  datetime64[ns]
 2   Gender                18590 non-null  object
 3   Company Type          18590 non-null  object
 4   WFH Setup Available   18590 non-null  object
 5   Designation           18590 non-null  int64
 6   Resource Allocation   18590 non-null  int64
 7   Mental Fatigue Score  18590 non-null  float64
 8   Burn Rate             18590 non-null  float64
dtypes: datetime64[ns](1), float64(2), int64(2), object(4)
memory usage: 1.3+ MB
```

[22]:
```
df.columns = [
    'employee_id',
    'date_of_joining',
    'gender',
    'company_type',
    'wfh_setup',
    'designation',
    'resource_allocation',
    'mental_fatigue_score',
    'burn_rate'
```

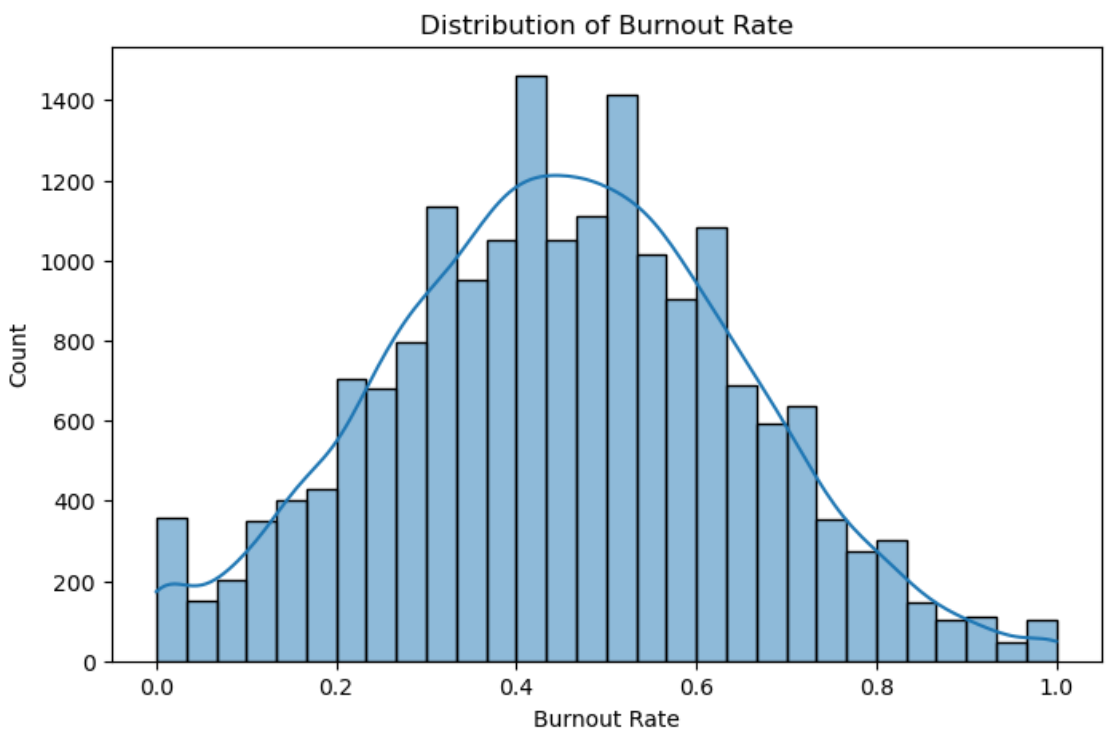
```
]
```

[23]:
```
df.columns
```

[23]:
```
Index(['employee_id', 'date_of_joining', 'gender', 'company_type', 'wfh_setup',
       'designation', 'resource_allocation', 'mental_fatigue_score',
       'burn_rate'],
      dtype='object')
```

[24]:
```
df.to_csv("Burnout_Phase2_Final.csv", index=False)
```

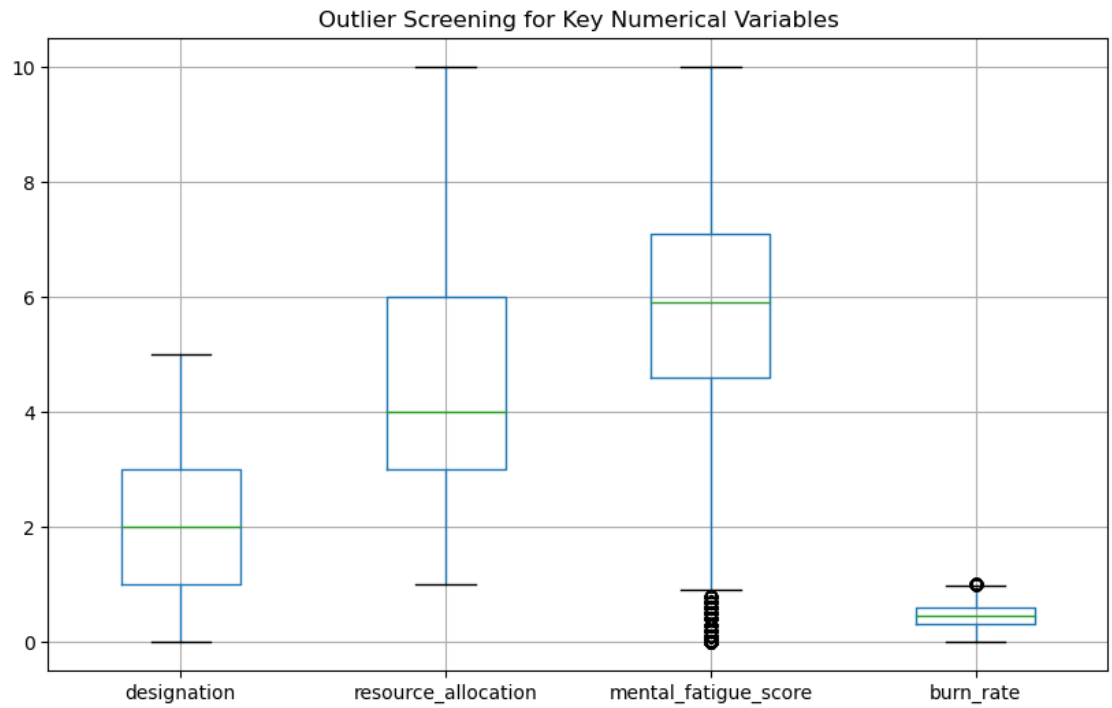
[25]:
```
plt.figure(figsize=(8,5))
sns.histplot(df['burn_rate'], bins=30, kde=True)
plt.title("Distribution of Burnout Rate")
plt.xlabel("Burnout Rate")
plt.ylabel("Count")
plt.show()

print("Burnout Rate Summary:")
print(df['burn_rate'].describe())
```



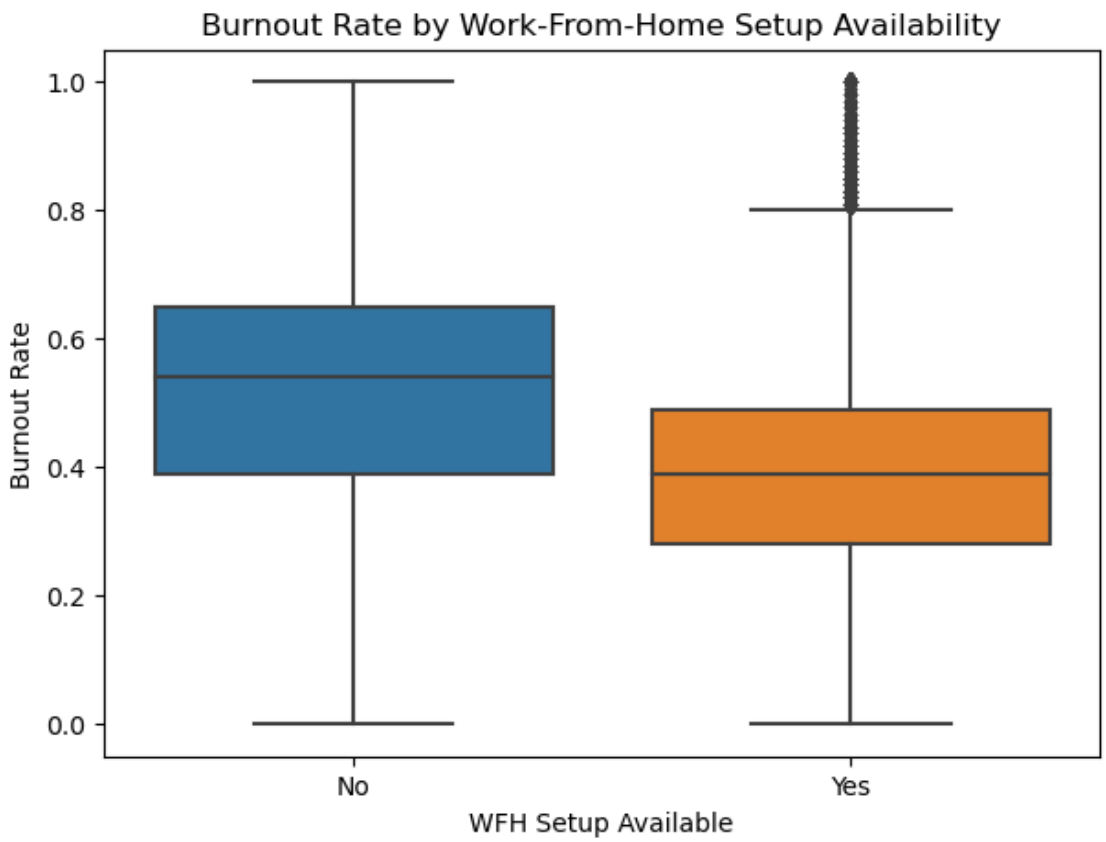

```
Burnout Rate Summary:
count    18590.000000
mean         0.452444
```

```
std          0.197848
min          0.000000
25%          0.320000
50%          0.450000
75%          0.590000
max          1.000000
Name: burn_rate, dtype: float64
```

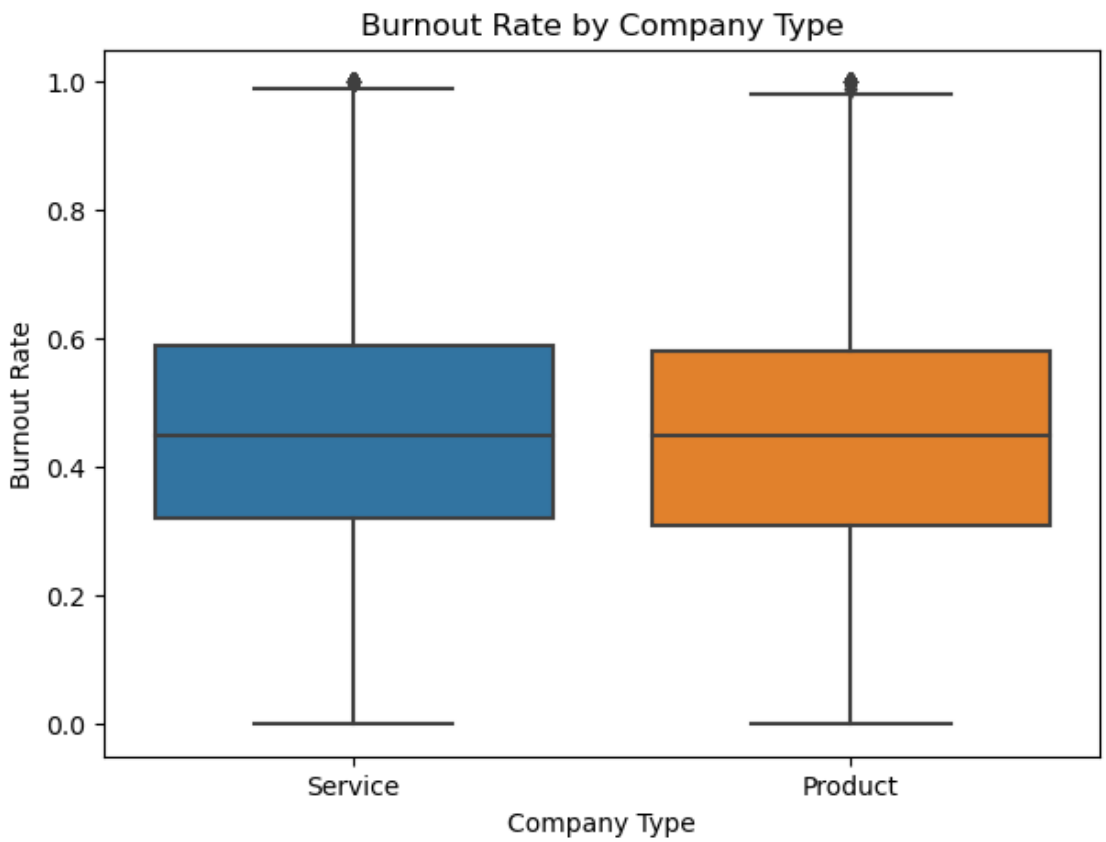
[26]:
```
num_cols = ['designation', 'resource_allocation', 'mental_fatigue_score',␣
↪'burn_rate']

plt.figure(figsize=(10,6))
df[num_cols].boxplot()
plt.title("Outlier Screening for Key Numerical Variables")
plt.show()
```



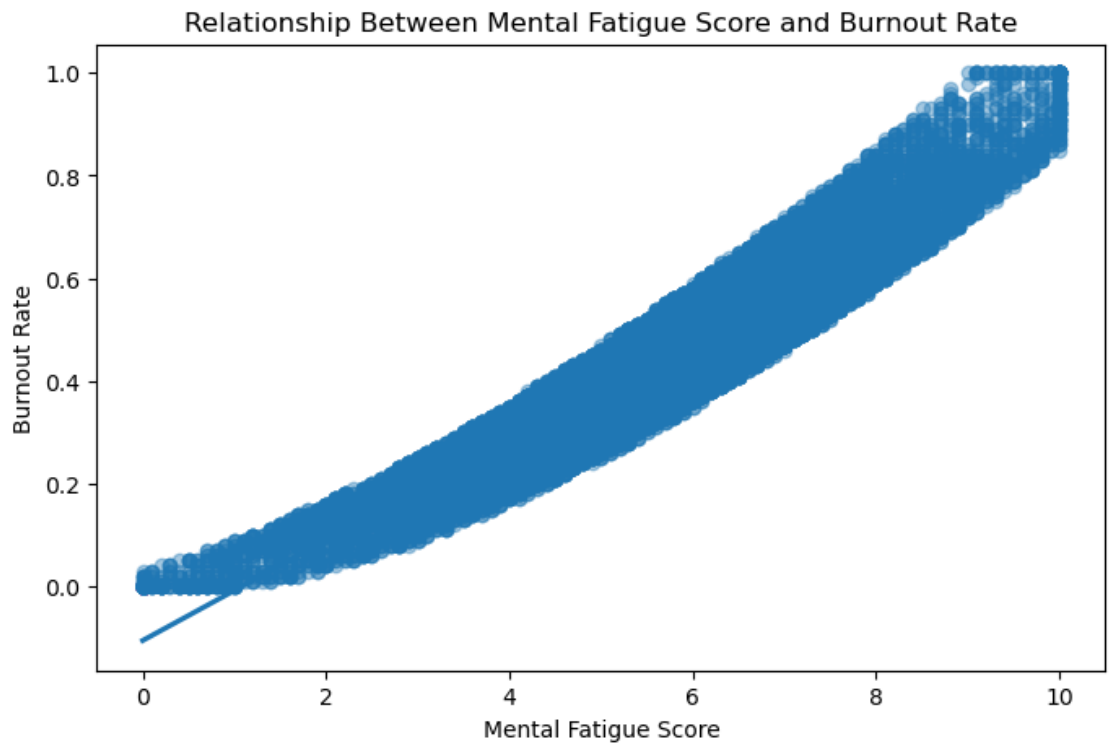
[27]:
```
plt.figure(figsize=(7,5))
sns.boxplot(x='wfh_setup', y='burn_rate', data=df)
plt.title("Burnout Rate by Work-From-Home Setup Availability")
plt.xlabel("WFH Setup Available")
plt.ylabel("Burnout Rate")
plt.show()
```



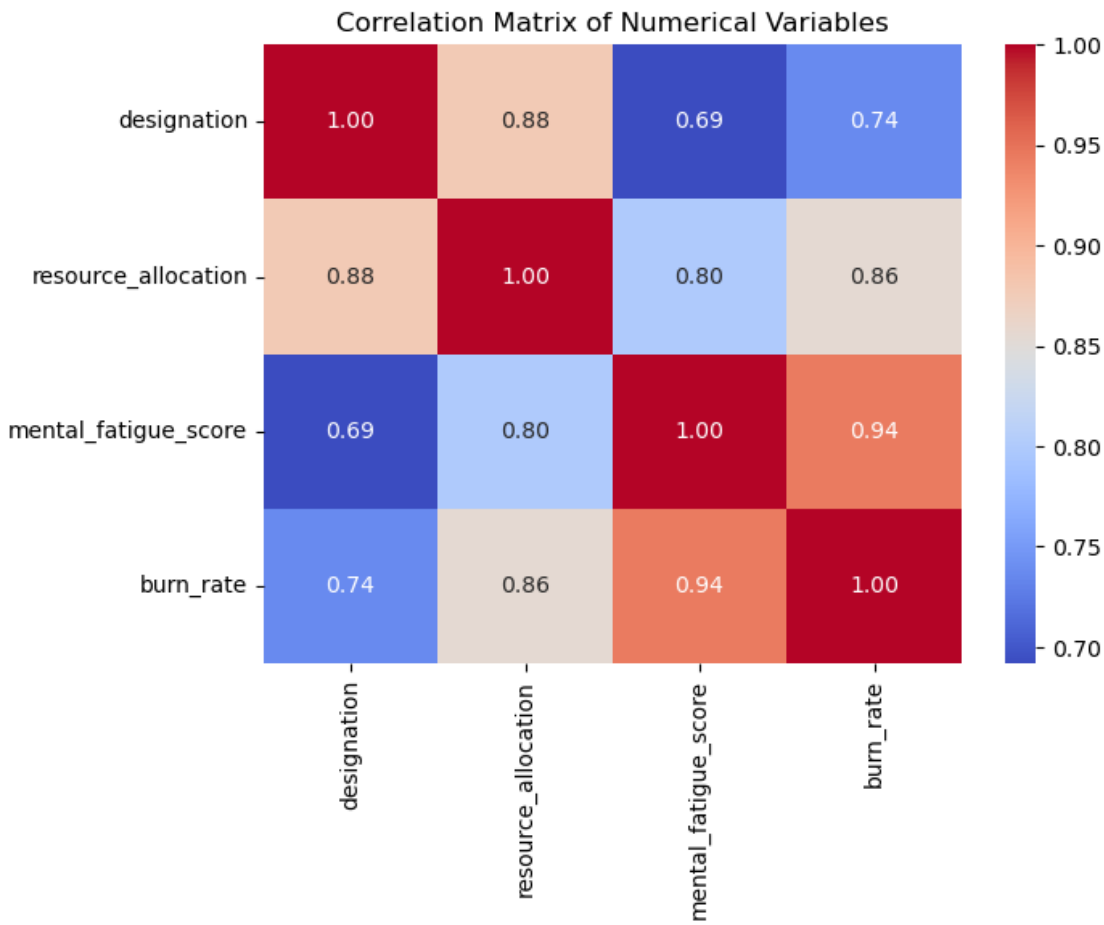
[28]:
```
plt.figure(figsize=(7,5))
sns.boxplot(x='company_type', y='burn_rate', data=df)
plt.title("Burnout Rate by Company Type")
plt.xlabel("Company Type")
plt.ylabel("Burnout Rate")
plt.show()
```



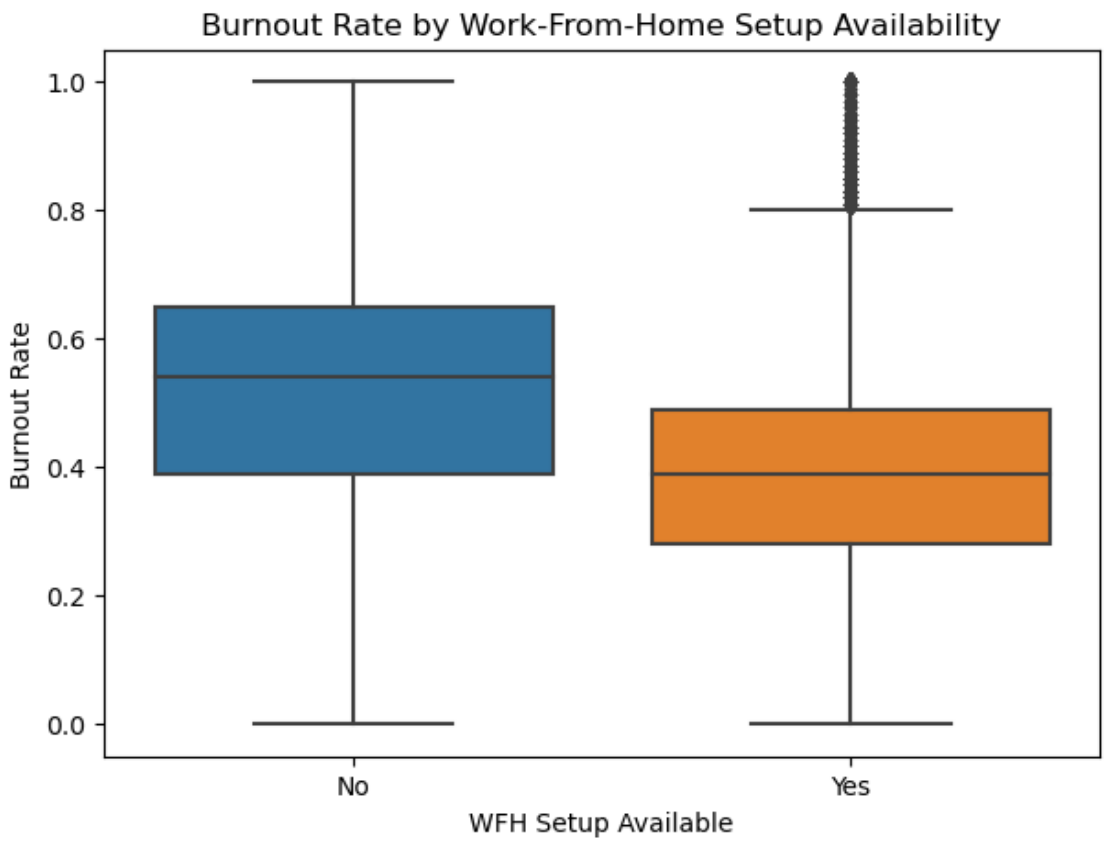
[29]:
```
plt.figure(figsize=(8,5))
sns.regplot(x='mental_fatigue_score', y='burn_rate', data=df,␣
↪scatter_kws={'alpha':0.4})
plt.title("Relationship Between Mental Fatigue Score and Burnout Rate")
plt.xlabel("Mental Fatigue Score")
plt.ylabel("Burnout Rate")
plt.show()
```



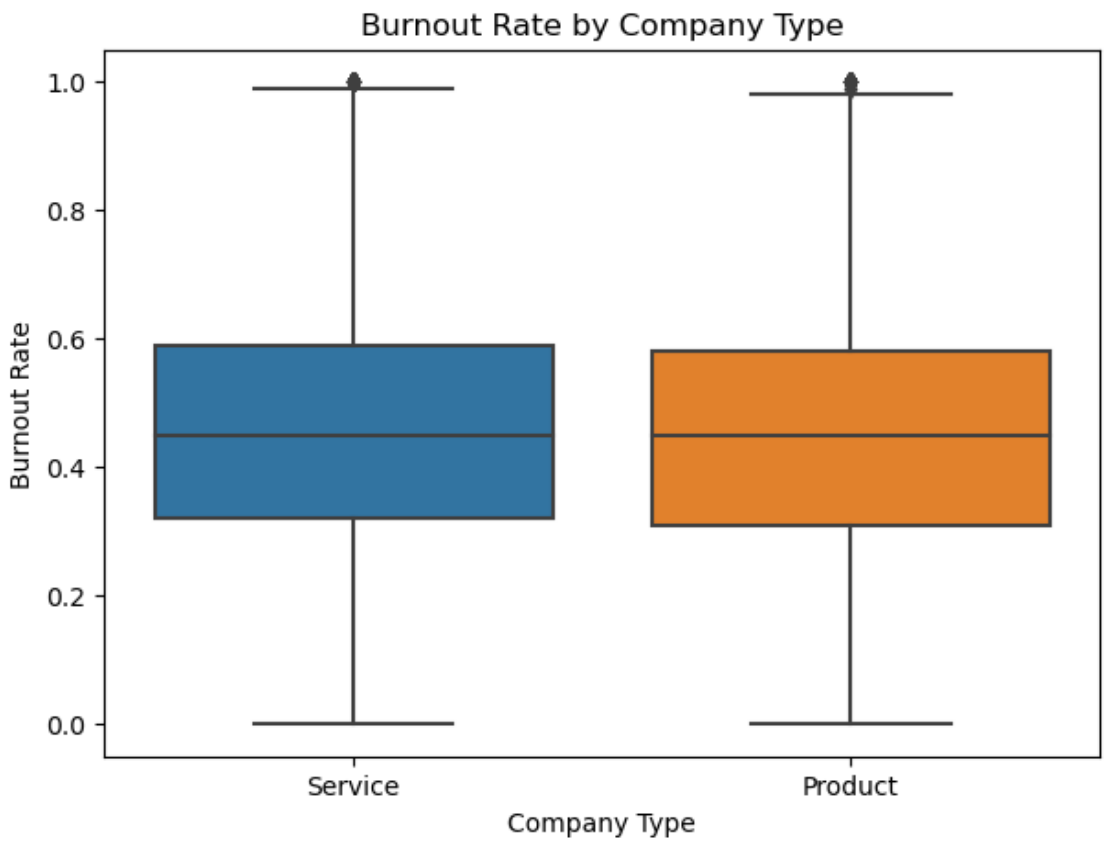
[30]:
```
corr = df[num_cols].corr()

plt.figure(figsize=(7,5))
sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Matrix of Numerical Variables")
plt.show()
```



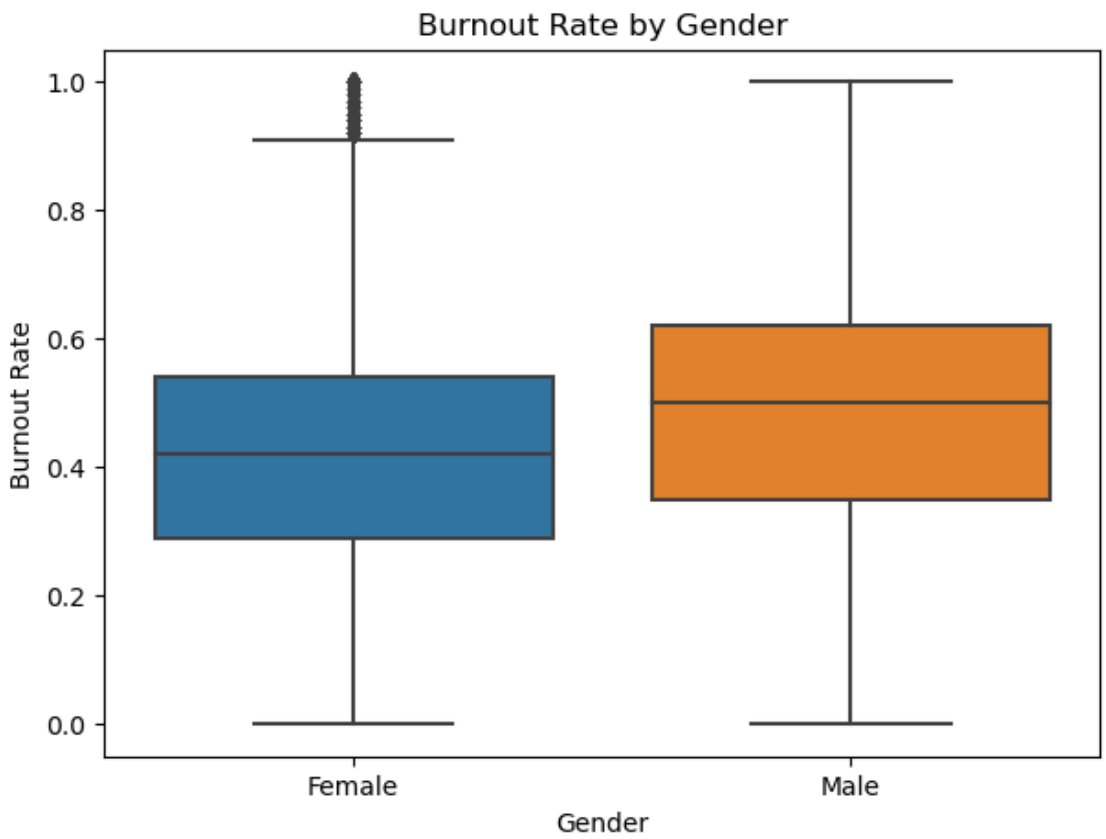
[31]:
```
plt.figure(figsize=(7,5))
sns.boxplot(x='wfh_setup', y='burn_rate', data=df)
plt.title("Burnout Rate by Work-From-Home Setup Availability")
plt.xlabel("WFH Setup Available")
plt.ylabel("Burnout Rate")
plt.show()
```



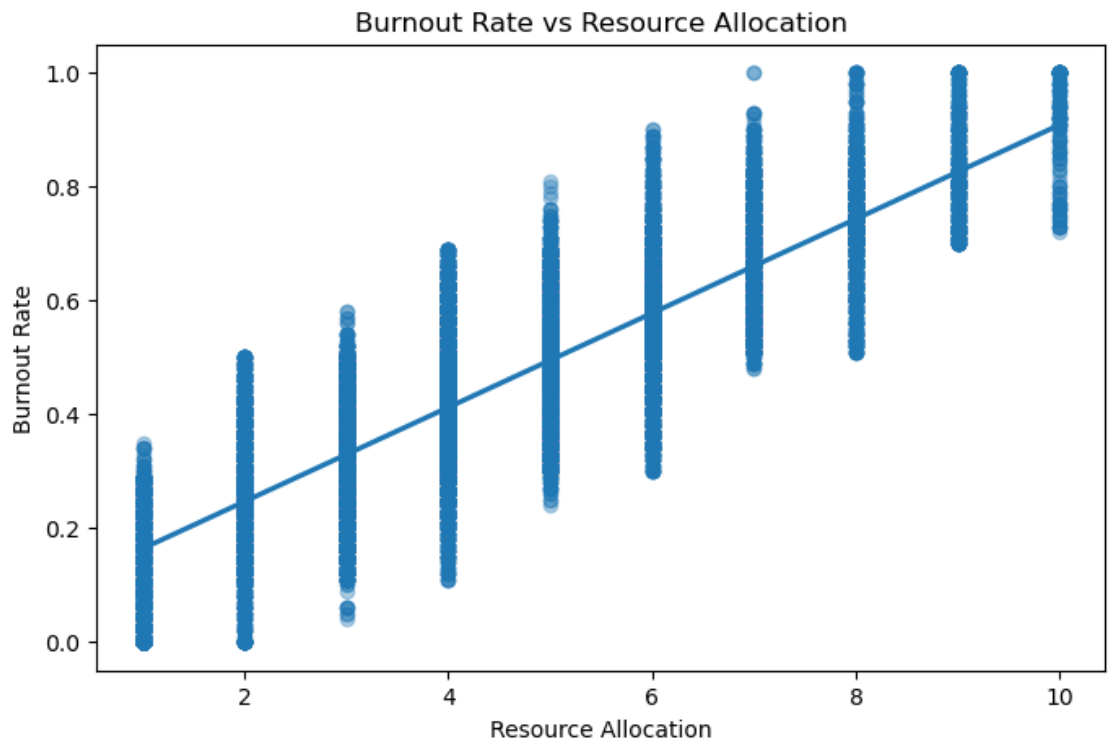
[32]:
```
plt.figure(figsize=(7,5))
sns.boxplot(x='company_type', y='burn_rate', data=df)
plt.title("Burnout Rate by Company Type")
plt.xlabel("Company Type")
plt.ylabel("Burnout Rate")
plt.show()
```



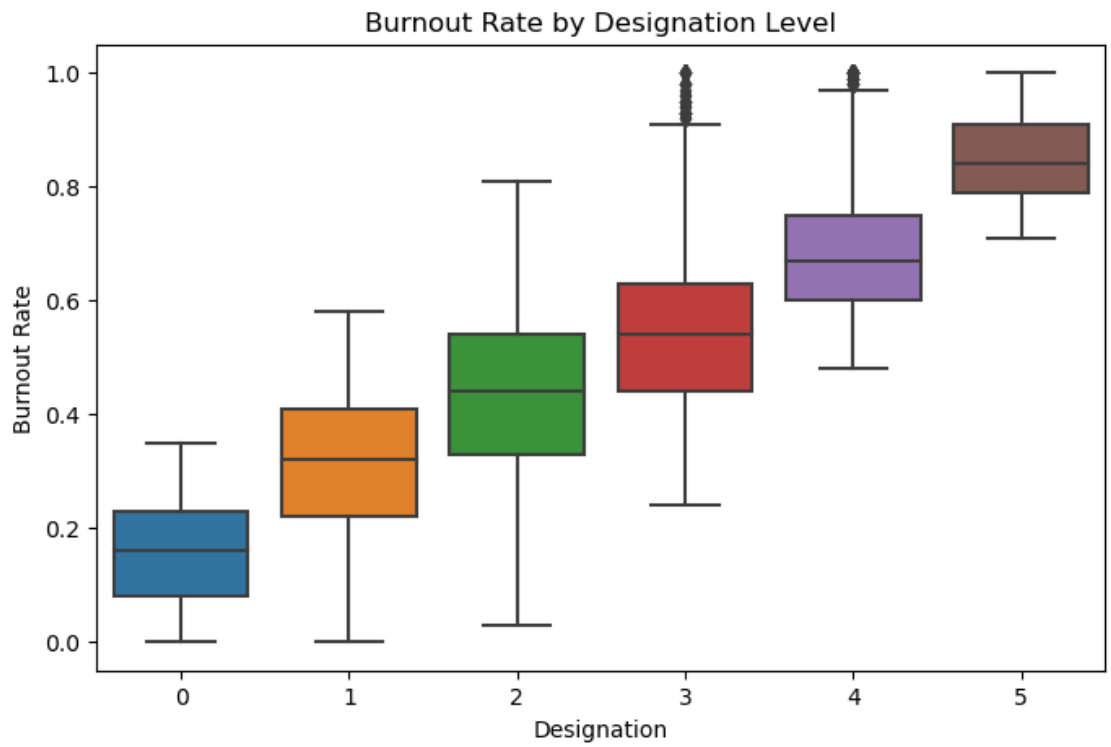
[33]:
```
plt.figure(figsize=(7,5))
sns.boxplot(x='gender', y='burn_rate', data=df)
plt.title("Burnout Rate by Gender")
plt.xlabel("Gender")
plt.ylabel("Burnout Rate")
plt.show()
```



[34]:
```
plt.figure(figsize=(8,5))
sns.regplot(x='resource_allocation', y='burn_rate', data=df,␣
↪scatter_kws={'alpha':0.35})
plt.title("Burnout Rate vs Resource Allocation")
plt.xlabel("Resource Allocation")
plt.ylabel("Burnout Rate")
plt.show()
```



[35]:
```
plt.figure(figsize=(8,5))
sns.boxplot(x='designation', y='burn_rate', data=df)
plt.title("Burnout Rate by Designation Level")
plt.xlabel("Designation")
plt.ylabel("Burnout Rate")
plt.show()
```

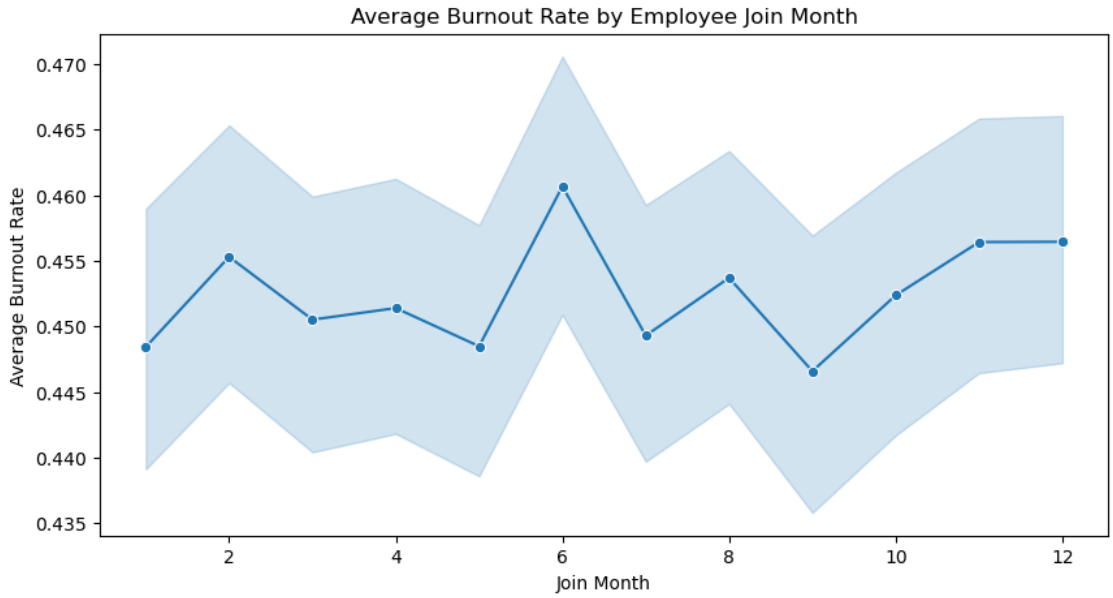


[36]:
```
df['join_year'] = df['date_of_joining'].dt.year
df['join_month'] = df['date_of_joining'].dt.month

plt.figure(figsize=(10,5))
sns.lineplot(x='join_year', y='burn_rate', data=df, estimator='mean',␣
 ↪errorbar=None)
plt.title("Average Burnout Rate by Employee Join Year")
plt.xlabel("Join Year")
plt.ylabel("Average Burnout Rate")
plt.show()
```

[37]:
```
df['join_year'].value_counts().sort_index()
```

[37]:
```
2008    18590
Name: join_year, dtype: int64
```

[38]:
```
df['join_month'] = df['date_of_joining'].dt.month

plt.figure(figsize=(10,5))
sns.lineplot(x='join_month', y='burn_rate', data=df, estimator='mean',␣
 ↪marker='o')
plt.title("Average Burnout Rate by Employee Join Month")
plt.xlabel("Join Month")
plt.ylabel("Average Burnout Rate")
plt.show()
```



[39]:
```
df['join_year'] = df['date_of_joining'].dt.year
df['join_month'] = df['date_of_joining'].dt.month
df['join_day'] = df['date_of_joining'].dt.day
df['join_weekday'] = df['date_of_joining'].dt.weekday
```

[40]:
```
df['fatigue_x_allocation'] = df['mental_fatigue_score'] *␣
↪df['resource_allocation']

df['fatigue_x_designation'] = df['mental_fatigue_score'] * df['designation']

df['allocation_x_designation'] = df['resource_allocation'] * df['designation']

df['workload_index'] = (
    df['mental_fatigue_score'] +
    df['resource_allocation'] +
    df['designation']
)
```

[41]:
```
df['fatigue_squared'] = df['mental_fatigue_score'] ** 2

df['allocation_squared'] = df['resource_allocation'] ** 2

df['designation_squared'] = df['designation'] ** 2
```

[42]:
```
df['fatigue_high_flag'] = (df['mental_fatigue_score'] >= 7).astype(int)

df['allocation_high_flag'] = (df['resource_allocation'] >= 7).astype(int)
```

```
df['senior_flag'] = (df['designation'] >= 4).astype(int)
```

[43]:
```
df_encoded = pd.get_dummies(df, columns=[
    'gender',
    'company_type',
    'wfh_setup'
], drop_first=True)
```

[44]:
```
df_encoded = df_encoded.drop(columns=[
    'employee_id',
    'date_of_joining'
])
```

[45]:
```
X = df_encoded.drop(columns=['burn_rate'])

y = df_encoded['burn_rate']
```

[46]:
```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

[47]:
```
print("Total features:", X.shape[1])
```

```
Total features: 20
```

[48]:
```
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor, GradientBoostingRegressor
import numpy as np
import pandas as pd
```

[49]:
```
def evaluate_regression(model, X_train, X_test, y_train, y_test,␣
 ↪model_name="Model"):
    model.fit(X_train, y_train)
    preds = model.predict(X_test)

    mae = mean_absolute_error(y_test, preds)
    rmse = np.sqrt(mean_squared_error(y_test, preds))
    r2 = r2_score(y_test, preds)

    return {
        "Model": model_name,
```

```
        "MAE": mae,
        "RMSE": rmse,
        "R2": r2
    }
```

[50]:
```
lr = LinearRegression()
lr_results = evaluate_regression(lr, X_train, X_test, y_train, y_test, "Linear␣
↪Regression")
lr_results
```

[50]:
```
{'Model': 'Linear Regression',
 'MAE': 0.04452085970384252,
 'RMSE': 0.05404382429099379,
 'R2': 0.9239486736455371}
```

[51]:
```
rf = RandomForestRegressor(
    n_estimators=300,
    random_state=42,
    max_depth=None,
    n_jobs=-1
)

rf_results = evaluate_regression(rf, X_train, X_test, y_train, y_test, "Random␣
↪Forest")
rf_results
```

[51]:
```
{'Model': 'Random Forest',
 'MAE': 0.04455688057668826,
 'RMSE': 0.0554770050437424,
 'R2': 0.9198616011906275}
```

[52]:
```
gbr = GradientBoostingRegressor(
    random_state=42,
    n_estimators=300,
    learning_rate=0.05,
    max_depth=3
)

gbr_results = evaluate_regression(gbr, X_train, X_test, y_train, y_test,␣
↪"Gradient Boosting")
gbr_results
```

[52]:
```
{'Model': 'Gradient Boosting',
 'MAE': 0.04327512798836826,
 'RMSE': 0.05286755018355568 4,
 'R2': 0.9272231894904852}
```

[53]:
```
results_df = pd.DataFrame([lr_results, rf_results, gbr_results])
results_df
```

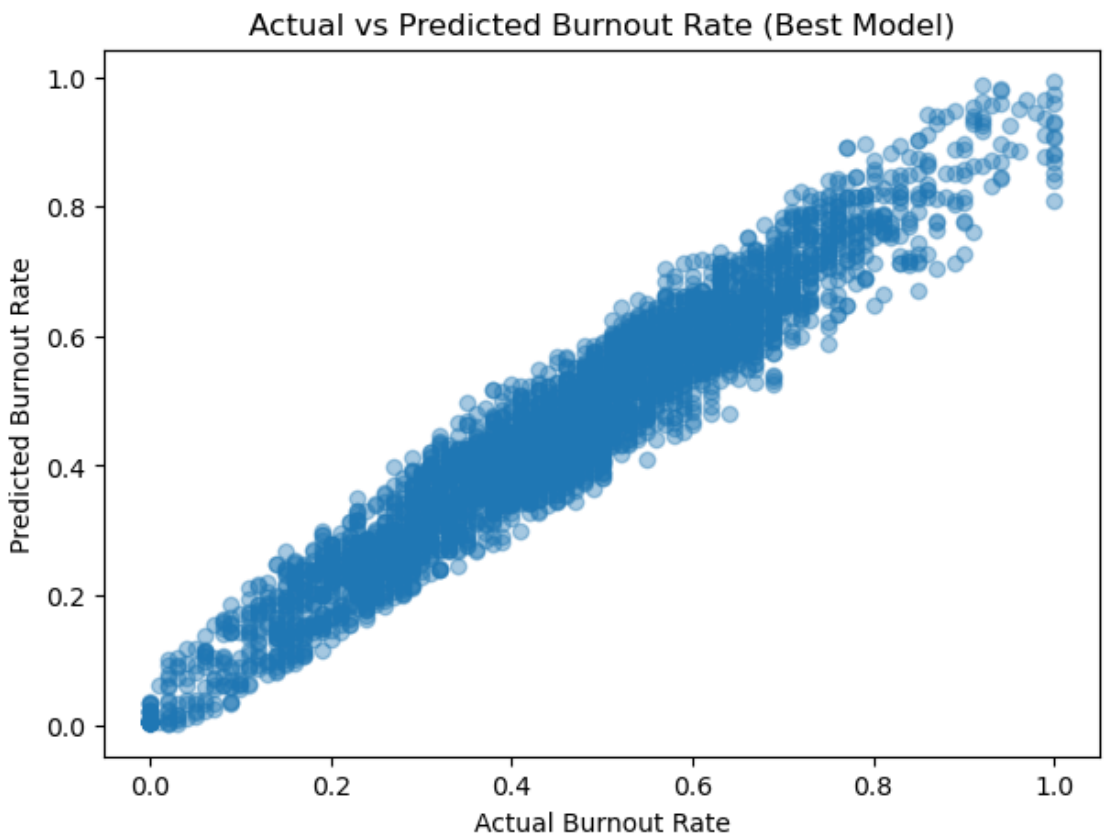
[53]:

	Model	MAE	RMSE	R2
0	Linear Regression	0.044521	0.054044	0.923949
1	Random Forest	0.044557	0.055477	0.919862
2	Gradient Boosting	0.043275	0.052868	0.927223

[54]:
```
best_model = results_df.sort_values("RMSE").iloc[0]
best_model
```

[54]:
```
Model    Gradient Boosting
MAE               0.043275
RMSE              0.052868
R2                0.927223
Name: 2, dtype: object
```

[56]:
```
best = gbr
best.fit(X_train, y_train)
preds = best.predict(X_test)

import matplotlib.pyplot as plt
plt.figure(figsize=(7,5))
plt.scatter(y_test, preds, alpha=0.4)
plt.title("Actual vs Predicted Burnout Rate (Best Model)")
plt.xlabel("Actual Burnout Rate")
plt.ylabel("Predicted Burnout Rate")
plt.show()
```



[57]:
```
import joblib
joblib.dump(best, "best_burnout_regression_model.pkl")
```

[57]: ['best_burnout_regression_model.pkl']

[58]:
```
def classify_burnout(rate):
    if rate < 0.33:
        return "Low"
    elif rate < 0.66:
        return "Medium"
    else:
        return "High"

df_encoded['burnout_risk'] = df_encoded['burn_rate'].apply(classify_burnout)

df_encoded['burnout_risk'].value_counts()
```

[58]:
```
Medium    10807
Low        4899
High       2884
```

```
Name: burnout_risk, dtype: int64
```

[59]:
```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()
df_encoded['burnout_risk_encoded'] = le.
 ↪fit_transform(df_encoded['burnout_risk'])
```

[60]:
```
X_class = df_encoded.drop(columns=['burn_rate', 'burnout_risk',␣
 ↪'burnout_risk_encoded'])
y_class = df_encoded['burnout_risk_encoded']
```

[61]:
```
X_train_c, X_test_c, y_train_c, y_test_c = train_test_split(
    X_class,
    y_class,
    test_size=0.2,
    random_state=42
)
```

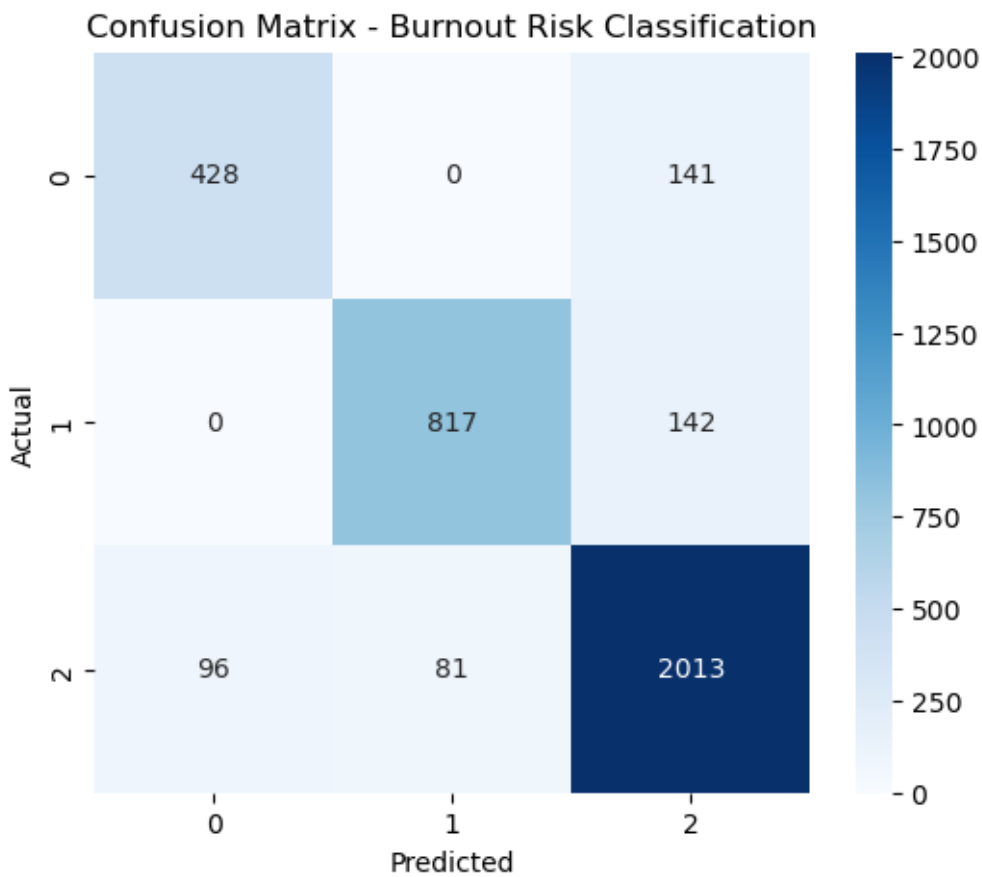
[62]:
```
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report,␣
 ↪confusion_matrix

clf = RandomForestClassifier(
    n_estimators=300,
    random_state=42
)

clf.fit(X_train_c, y_train_c)

preds_c = clf.predict(X_test_c)
```

[63]:
```
accuracy = accuracy_score(y_test_c, preds_c)
print("Accuracy:", accuracy)

print("\nClassification Report:")
print(classification_report(y_test_c, preds_c))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test_c, preds_c))
```

```
Accuracy: 0.8762775685852608

Classification Report:
              precision    recall  f1-score   support

           0       0.82      0.75      0.78       569
           1       0.91      0.85      0.88       959
```

```
           2       0.88      0.92      0.90      2190

    accuracy                           0.88      3718
   macro avg       0.87      0.84      0.85      3718
weighted avg       0.88      0.88      0.88      3718


Confusion Matrix:
[[ 428    0  141]
 [   0  817  142]
 [  96   81 2013]]
```

[64]:
```python
import seaborn as sns
import matplotlib.pyplot as plt

cm = confusion_matrix(y_test_c, preds_c)

plt.figure(figsize=(6,5))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues")
plt.title("Confusion Matrix - Burnout Risk Classification")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```



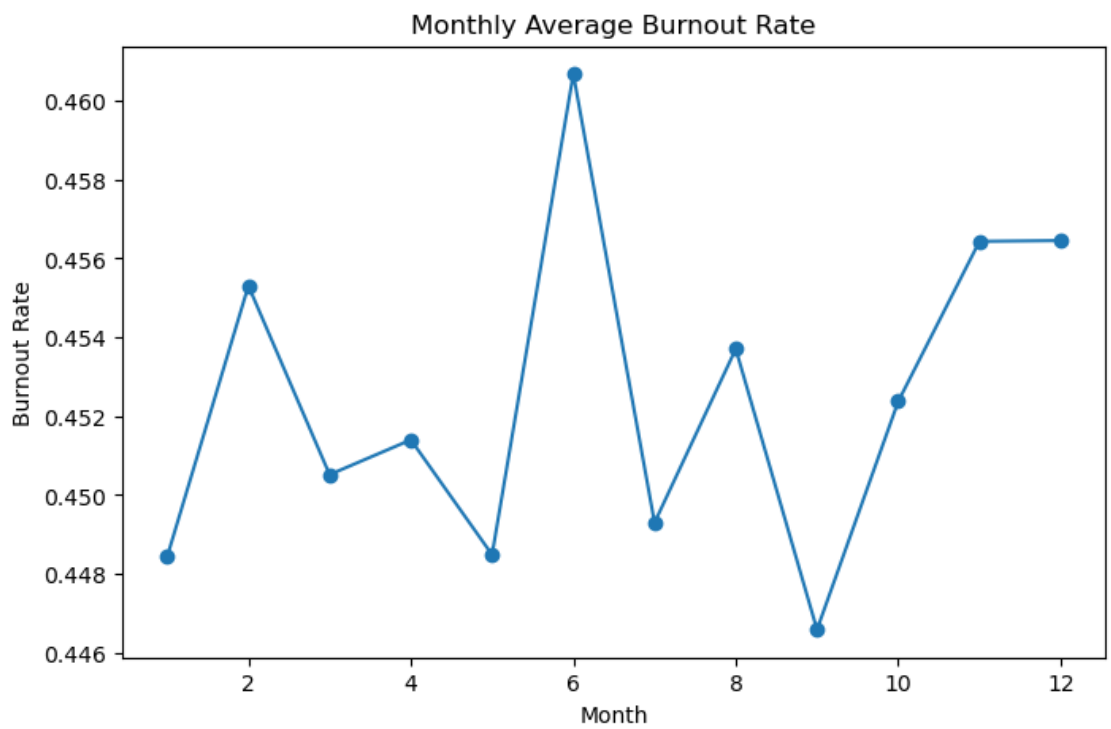
[65]:
```
ts = df.groupby('join_month')['burn_rate'].mean().reset_index()

ts.columns = ['month', 'burn_rate']

ts
```

[65]:

	month	burn_rate
0	1	0.448433
1	2	0.455306
2	3	0.450516
3	4	0.451399
4	5	0.448485
..	…	…
7	8	0.453705
8	9	0.446584
9	10	0.452373
10	11	0.456434
11	12	0.456453

```
[12 rows x 2 columns]
```

[66]:
```
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))
plt.plot(ts['month'], ts['burn_rate'], marker='o')
plt.title("Monthly Average Burnout Rate")
plt.xlabel("Month")
plt.ylabel("Burnout Rate")
plt.show()
```



[67]:
```
pip install prophet
```

```
Requirement already satisfied: prophet in c:\users\rpere\anaconda3\lib\site-
packages (1.3.0)Note: you may need to restart the kernel to use updated
packages.

Requirement already satisfied: cmdstanpy>=1.0.4 in
c:\users\rpere\anaconda3\lib\site-packages (from prophet) (1.3.0)
Requirement already satisfied: numpy>=1.15.4 in
c:\users\rpere\anaconda3\lib\site-packages (from prophet) (1.24.3)
Requirement already satisfied: matplotlib>=2.0.0 in
c:\users\rpere\anaconda3\lib\site-packages (from prophet) (3.7.1)
Requirement already satisfied: pandas>=1.0.4 in
```

```
c:\users\rpere\anaconda3\lib\site-packages (from prophet) (1.5.3)
Requirement already satisfied: holidays<1,>=0.25 in
c:\users\rpere\anaconda3\lib\site-packages (from prophet) (0.91)
Requirement already satisfied: tqdm>=4.36.1 in
c:\users\rpere\anaconda3\lib\site-packages (from prophet) (4.65.0)
Requirement already satisfied: importlib_resources in
c:\users\rpere\anaconda3\lib\site-packages (from prophet) (6.5.2)
Requirement already satisfied: stanio<2.0.0,>=0.4.0 in
c:\users\rpere\anaconda3\lib\site-packages (from cmdstanpy>=1.0.4->prophet)
(0.5.1)
Requirement already satisfied: python-dateutil<3,>=2.9.0.post0 in
c:\users\rpere\anaconda3\lib\site-packages (from holidays<1,>=0.25->prophet)
(2.9.0.post0)
Requirement already satisfied: contourpy>=1.0.1 in
c:\users\rpere\anaconda3\lib\site-packages (from matplotlib>=2.0.0->prophet)
(1.0.5)
Requirement already satisfied: cycler>=0.10 in
c:\users\rpere\anaconda3\lib\site-packages (from matplotlib>=2.0.0->prophet)
(0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in
c:\users\rpere\anaconda3\lib\site-packages (from matplotlib>=2.0.0->prophet)
(4.25.0)
Requirement already satisfied: kiwisolver>=1.0.1 in
c:\users\rpere\anaconda3\lib\site-packages (from matplotlib>=2.0.0->prophet)
(1.4.4)
Requirement already satisfied: packaging>=20.0 in
c:\users\rpere\anaconda3\lib\site-packages (from matplotlib>=2.0.0->prophet)
(23.0)
Requirement already satisfied: pillow>=6.2.0 in
c:\users\rpere\anaconda3\lib\site-packages (from matplotlib>=2.0.0->prophet)
(9.4.0)
Requirement already satisfied: pyparsing>=2.3.1 in
c:\users\rpere\anaconda3\lib\site-packages (from matplotlib>=2.0.0->prophet)
(3.0.9)
Requirement already satisfied: pytz>=2020.1 in
c:\users\rpere\anaconda3\lib\site-packages (from pandas>=1.0.4->prophet)
(2022.7)
Requirement already satisfied: colorama in c:\users\rpere\anaconda3\lib\site-
packages (from tqdm>=4.36.1->prophet) (0.4.6)
Requirement already satisfied: six>=1.5 in c:\users\rpere\anaconda3\lib\site-
packages (from python-dateutil<3,>=2.9.0.post0->holidays<1,>=0.25->prophet)
(1.16.0)
```

[68]:
```python
from prophet import Prophet

ts_prophet = ts.copy()
```

```
ts_prophet['ds'] = pd.date_range(start='2008-01-01', periods=len(ts_prophet),␣
↪freq='M')

ts_prophet['y'] = ts_prophet['burn_rate']

ts_prophet = ts_prophet[['ds','y']]

ts_prophet
```

[68]:

```
           ds         y
0  2008-01-31  0.448433
1  2008-02-29  0.455306
2  2008-03-31  0.450516
3  2008-04-30  0.451399
4  2008-05-31  0.448485
..        …       …
7  2008-08-31  0.453705
8  2008-09-30  0.446584
9  2008-10-31  0.452373
10 2008-11-30  0.456434
11 2008-12-31  0.456453

[12 rows x 2 columns]
```

[69]:

```
model = Prophet()

model.fit(ts_prophet)
```

```
19:05:25 - cmdstanpy - INFO - Chain [1] start processing
19:05:26 - cmdstanpy - INFO - Chain [1] done processing
```

[69]:

```
<prophet.forecaster.Prophet at 0x1b8802a8390>
```

[71]:

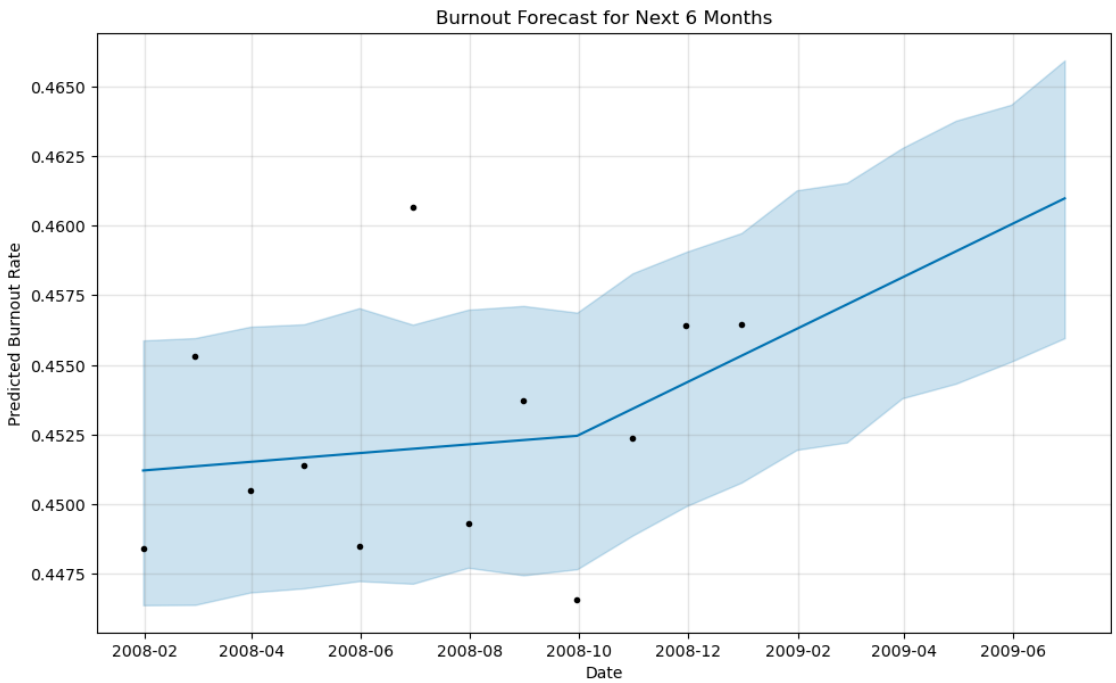
```
future = model.make_future_dataframe(periods=6, freq='M')
forecast = model.predict(future)

forecast[['ds','yhat','yhat_lower','yhat_upper']]
```

[71]:

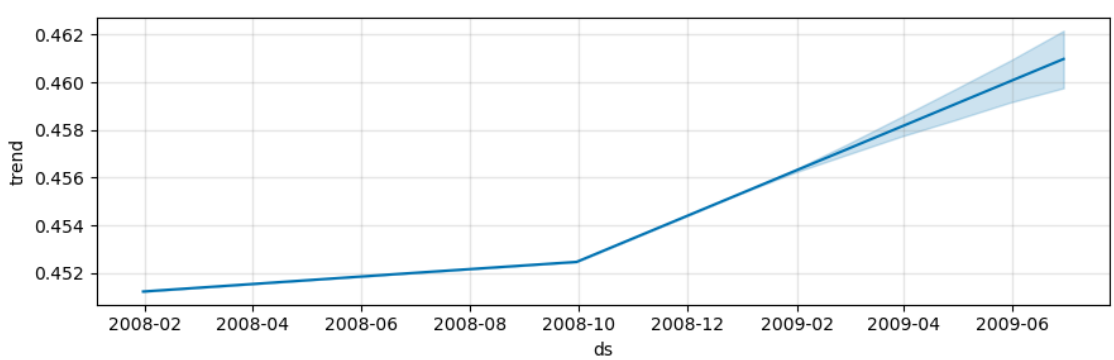
```
           ds      yhat  yhat_lower  yhat_upper
0  2008-01-31  0.451216    0.446384    0.455884
1  2008-02-29  0.451365    0.446394    0.455969
2  2008-03-31  0.451525    0.446839    0.456376
3  2008-04-30  0.451680    0.446988    0.456463
4  2008-05-31  0.451840    0.447248    0.457043
..        …       …           …          …
13 2009-02-28  0.457173    0.452224    0.461538
14 2009-03-31  0.458140    0.453807    0.462789
15 2009-04-30  0.459077    0.454332    0.463768
```

```
16 2009-05-31  0.460044    0.455122    0.464342
17 2009-06-30  0.460980    0.455966    0.465933

[18 rows x 4 columns]
```

[72]:
```
fig = model.plot(forecast)
plt.title("Burnout Forecast for Next 6 Months")
plt.xlabel("Date")
plt.ylabel("Predicted Burnout Rate")
plt.show()
```



[73]:
```
fig2 = model.plot_components(forecast)
plt.show()
```



[74]:
```
future_forecast = forecast.tail(6)

future_forecast[['ds','yhat','yhat_lower','yhat_upper']]
```

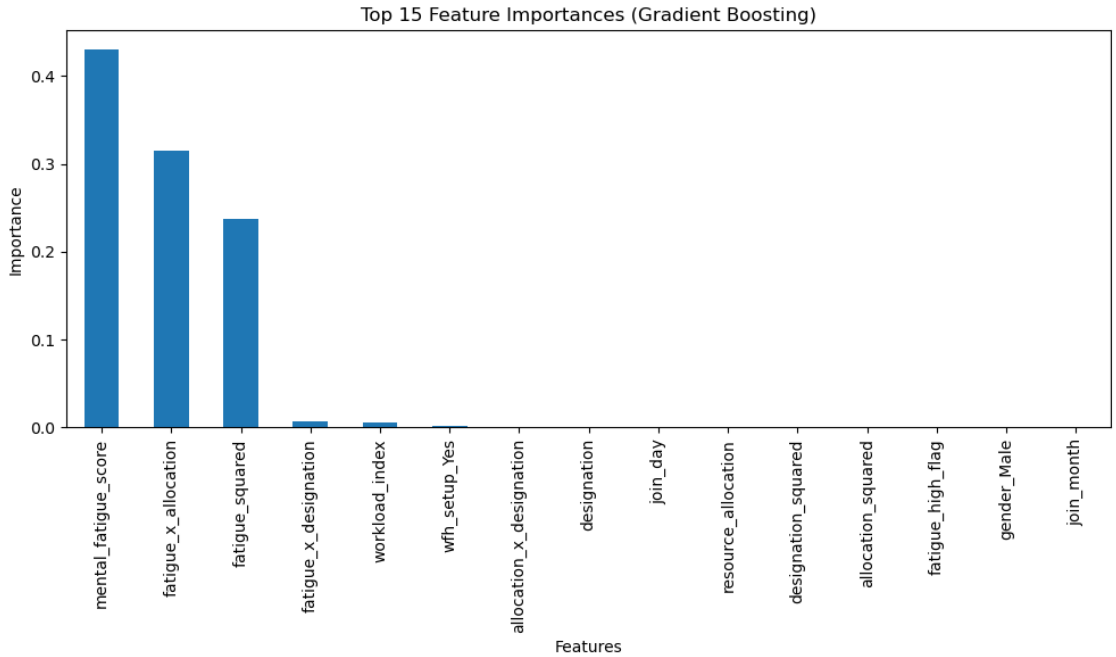
[74]:

	ds	yhat	yhat_lower	yhat_upper
12	2009-01-31	0.456299	0.451959	0.461271
13	2009-02-28	0.457173	0.452224	0.461538
14	2009-03-31	0.458140	0.453807	0.462789
15	2009-04-30	0.459077	0.454332	0.463768
16	2009-05-31	0.460044	0.455122	0.464342
17	2009-06-30	0.460980	0.455966	0.465933

[75]:
```
from sklearn.ensemble import GradientBoostingRegressor

best = GradientBoostingRegressor(
    random_state=42,
    n_estimators=300,
    learning_rate=0.05,
    max_depth=3
)
best.fit(X_train, y_train)
```

[75]:
```
GradientBoostingRegressor(learning_rate=0.05, n_estimators=300, random_state=42)
```

[76]:
```
import pandas as pd
import matplotlib.pyplot as plt

importances = pd.Series(best.feature_importances_, index=X_train.columns).
 ↪sort_values(ascending=False)

plt.figure(figsize=(10,6))
importances.head(15).plot(kind="bar")
plt.title("Top 15 Feature Importances (Gradient Boosting)")
plt.xlabel("Features")
plt.ylabel("Importance")
plt.tight_layout()
plt.show()

importances.head(20)
```



[76]:
```
mental_fatigue_score     0.430139
fatigue_x_allocation     0.314587
fatigue_squared          0.237876
fatigue_x_designation    0.007296
workload_index           0.006161
                           …
join_weekday             0.000069
company_type_Service     0.000056
senior_flag              0.000048
allocation_high_flag     0.000017
join_year                0.000000
Length: 20, dtype: float64
```

[77]:
```
pip install shap
```

```
Collecting shap
  Obtaining dependency information for shap from https://files.pythonhosted.org/
packages/77/03/58e199cf59056d68b4a227ce4b2b09eeb0c9bd1d002b9e28fb574eed6200/shap
-0.50.0-cp311-cp311-win_amd64.whl.metadata
  Downloading shap-0.50.0-cp311-cp311-win_amd64.whl.metadata (25 kB)
Collecting numpy>=2 (from shap)
  Obtaining dependency information for numpy>=2 from https://files.pythonhosted.
org/packages/76/ae/e0265e0163cf127c24c3969d29f1c4c64551a1e375d95a13d32eab25d364/
numpy-2.4.2-cp311-cp311-win_amd64.whl.metadata
  Downloading numpy-2.4.2-cp311-cp311-win_amd64.whl.metadata (6.6 kB)
Requirement already satisfied: scipy in c:\users\rpere\anaconda3\lib\site-
```

```
packages (from shap) (1.10.1)
Requirement already satisfied: scikit-learn in
c:\users\rpere\anaconda3\lib\site-packages (from shap) (1.3.2)
Requirement already satisfied: pandas in c:\users\rpere\anaconda3\lib\site-
packages (from shap) (1.5.3)
Requirement already satisfied: tqdm>=4.27.0 in
c:\users\rpere\anaconda3\lib\site-packages (from shap) (4.65.0)
Requirement already satisfied: packaging>20.9 in
c:\users\rpere\anaconda3\lib\site-packages (from shap) (23.0)
Collecting slicer==0.0.8 (from shap)
  Obtaining dependency information for slicer==0.0.8 from https://files.pythonho
sted.org/packages/63/81/9ef641ff4e12cbcca30e54e72fb0951a2ba195d0cda0ba4100e532d9
29db/slicer-0.0.8-py3-none-any.whl.metadata
  Downloading slicer-0.0.8-py3-none-any.whl.metadata (4.0 kB)
Requirement already satisfied: numba>=0.54 in c:\users\rpere\anaconda3\lib\site-
packages (from shap) (0.57.0)
Requirement already satisfied: cloudpickle in c:\users\rpere\anaconda3\lib\site-
packages (from shap) (2.2.1)
Requirement already satisfied: typing-extensions in
c:\users\rpere\anaconda3\lib\site-packages (from shap) (4.15.0)
Requirement already satisfied: llvmlite<0.41,>=0.40.0dev0 in
c:\users\rpere\anaconda3\lib\site-packages (from numba>=0.54->shap) (0.40.0)
INFO: pip is looking at multiple versions of numba to determine which version is
compatible with other requirements. This could take a while.
Collecting numba>=0.54 (from shap)
  Obtaining dependency information for numba>=0.54 from https://files.pythonhost
ed.org/packages/31/29/e09d5630578a50a2b3fa154990b6b839cf95327aa0709e2d50d0b6816c
d1/numba-0.64.0-cp311-cp311-win_amd64.whl.metadata
  Downloading numba-0.64.0-cp311-cp311-win_amd64.whl.metadata (2.8 kB)
Collecting llvmlite<0.47,>=0.46.0dev0 (from numba>=0.54->shap)
  Obtaining dependency information for llvmlite<0.47,>=0.46.0dev0 from https://f
iles.pythonhosted.org/packages/19/0c/8f5a37a65fc9b7b17408508145edd5f86263ad69c19
d3574e818f533a0eb/llvmlite-0.46.0-cp311-cp311-win_amd64.whl.metadata
  Downloading llvmlite-0.46.0-cp311-cp311-win_amd64.whl.metadata (4.9 kB)
Requirement already satisfied: colorama in c:\users\rpere\anaconda3\lib\site-
packages (from tqdm>=4.27.0->shap) (0.4.6)
Requirement already satisfied: python-dateutil>=2.8.1 in
c:\users\rpere\anaconda3\lib\site-packages (from pandas->shap) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
c:\users\rpere\anaconda3\lib\site-packages (from pandas->shap) (2022.7)
INFO: pip is looking at multiple versions of scikit-learn to determine which
version is compatible with other requirements. This could take a while.
Collecting scikit-learn (from shap)
  Obtaining dependency information for scikit-learn from https://files.pythonhos
ted.org/packages/89/3c/45c352094cfa60050bcbb967b1faf246b22e93cb459f2f907b600f2ce
da5/scikit_learn-1.8.0-cp311-cp311-win_amd64.whl.metadata
  Using cached scikit_learn-1.8.0-cp311-cp311-win_amd64.whl.metadata (11 kB)
Requirement already satisfied: joblib>=1.3.0 in
```

```
c:\users\rpere\anaconda3\lib\site-packages (from scikit-learn->shap) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.2.0 in
c:\users\rpere\anaconda3\lib\site-packages (from scikit-learn->shap) (3.6.0)
INFO: pip is looking at multiple versions of scipy to determine which version is
compatible with other requirements. This could take a while.
Collecting scipy (from shap)
  Obtaining dependency information for scipy from https://files.pythonhosted.org
/packages/52/c8/08629657ac6c0da198487ce8cd3de78e02cfde42b7f34117d56a3fe249dc/sci
py-1.17.0-cp311-cp311-win_amd64.whl.metadata
  Downloading scipy-1.17.0-cp311-cp311-win_amd64.whl.metadata (60 kB)
     ---------------------------------------- 0.0/61.0 kB ? eta -:--:--
     -------------------- ------------------- 30.7/61.0 kB ? eta -:--:--
     -------------------- ------------------- 30.7/61.0 kB ? eta -:--:--
     -------------------- ------------------- 30.7/61.0 kB ? eta -:--:--
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     -------------------- ------------------- 30.7/61.0 kB ? eta -:--:--
     -------------------- ------------------- 30.7/61.0 kB ? eta -:--:--
     -------------------------- ------------ 41.0/61.0 kB 26.9 kB/s eta 0:00:01
     -------------------------------- ------ 51.2/61.0 kB 34.5 kB/s eta 0:00:01
     -------------------------------- ------ 51.2/61.0 kB 34.5 kB/s eta 0:00:01
     -------------------------------- ------ 51.2/61.0 kB 34.5 kB/s eta 0:00:01
     -------------------------------- ------ 51.2/61.0 kB 34.5 kB/s eta 0:00:01
     ---------------------------------------- 61.0/61.0 kB 37.3 kB/s eta 0:00:00
Requirement already satisfied: six>=1.5 in c:\users\rpere\anaconda3\lib\site-
packages (from python-dateutil>=2.8.1->pandas->shap) (1.16.0)
Downloading shap-0.50.0-cp311-cp311-win_amd64.whl (547 kB)
   ---------------------------------------- 0.0/548.0 kB ? eta -:--:--
   -- ------------------------------------- 30.7/548.0 kB ? eta -:--:--
   ------ --------------------------------- 92.2/548.0 kB 1.3 MB/s eta 0:00:01
   ------------ --------------------------- 174.1/548.0 kB 1.5 MB/s eta 0:00:01
   ------------------ --------------------- 256.0/548.0 kB 1.6 MB/s eta 0:00:01
```

```
   -------------------------- ------------- 358.4/548.0 kB 1.7 MB/s eta 0:00:01
   ------------------------------ -------- 430.1/548.0 kB 1.7 MB/s eta 0:00:01
   ------------------------------------- - 532.5/548.0 kB 1.8 MB/s eta 0:00:01
   -------------------------------------- 548.0/548.0 kB 1.6 MB/s eta 0:00:00
Downloading slicer-0.0.8-py3-none-any.whl (15 kB)
Downloading numba-0.64.0-cp311-cp311-win_amd64.whl (2.7 MB)
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learn, shap
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    Uninstalling numpy-1.24.3:
      Successfully uninstalled numpy-1.24.3
  Attempting uninstall: llvmlite
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    Uninstalling llvmlite-0.40.0:
      Successfully uninstalled llvmlite-0.40.0
  Attempting uninstall: scipy
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    Uninstalling scipy-1.10.1:
      Successfully uninstalled scipy-1.10.1
  Attempting uninstall: numba
    Found existing installation: numba 0.57.0
    Uninstalling numba-0.57.0:
      Successfully uninstalled numba-0.57.0
  Attempting uninstall: scikit-learn
    Found existing installation: scikit-learn 1.3.2
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    Uninstalling scikit-learn-1.3.2:
      Successfully uninstalled scikit-learn-1.3.2
Note: you may need to restart the kernel to use updated packages.

ERROR: Could not install packages due to an OSError: [WinError 5] Access is
denied: 'C:\\Users\\rpere\\anaconda3\\Lib\\site-
packages\\~.learn\\.libs\\msvcp140.dll'
Consider using the `--user` option or check the permissions.
```

[78]:
```
import shap
shap.initjs()

# Use TreeExplainer for tree models
explainer = shap.Explainer(best, X_train)
shap_values = explainer(X_test)

# Summary plot (global)
shap.summary_plot(shap_values, X_test, show=True)
```

```
---------------------------------------------------------------------------
ModuleNotFoundError                       Traceback (most recent call last)
Cell In[78], line 1
----> 1 import shap
      2 shap.initjs()
      4 # Use TreeExplainer for tree models

ModuleNotFoundError: No module named 'shap'
```

[ ]:
```
shap.summary_plot(shap_values, X_test, plot_type="bar", show=True)
```

[ ]:
```
i = 1 # change to 1,2,3 if you want another employee
shap.plots.waterfall(shap_values[1])
```

[79]:
```
import sys
!{sys.executable} -m pip install shap
```

```
Collecting shap
  Obtaining dependency information for shap from https://files.pythonhosted.org/
packages/77/03/58e199cf59056d68b4a227ce4b2b09eeb0c9bd1d002b9e28fb574eed6200/shap
-0.50.0-cp311-cp311-win_amd64.whl.metadata
  Using cached shap-0.50.0-cp311-cp311-win_amd64.whl.metadata (25 kB)
Requirement already satisfied: numpy>=2 in c:\users\rpere\anaconda3\lib\site-
packages (from shap) (2.4.2)
Requirement already satisfied: scipy in c:\users\rpere\anaconda3\lib\site-
packages (from shap) (1.17.0)
Requirement already satisfied: scikit-learn in
c:\users\rpere\anaconda3\lib\site-packages (from shap) (1.8.0)
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c:\users\rpere\anaconda3\lib\site-packages (from shap) (4.65.0)
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c:\users\rpere\anaconda3\lib\site-packages (from shap) (23.0)
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c:\users\rpere\anaconda3\lib\site-packages (from shap) (4.15.0)
Requirement already satisfied: llvmlite<0.47,>=0.46.0dev0 in
c:\users\rpere\anaconda3\lib\site-packages (from numba>=0.54->shap) (0.46.0)
Requirement already satisfied: colorama in c:\users\rpere\anaconda3\lib\site-
packages (from tqdm>=4.27.0->shap) (0.4.6)
Requirement already satisfied: python-dateutil>=2.8.1 in
c:\users\rpere\anaconda3\lib\site-packages (from pandas->shap) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
c:\users\rpere\anaconda3\lib\site-packages (from pandas->shap) (2022.7)
Requirement already satisfied: joblib>=1.3.0 in
c:\users\rpere\anaconda3\lib\site-packages (from scikit-learn->shap) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.2.0 in
c:\users\rpere\anaconda3\lib\site-packages (from scikit-learn->shap) (3.6.0)
Requirement already satisfied: six>=1.5 in c:\users\rpere\anaconda3\lib\site-
packages (from python-dateutil>=2.8.1->pandas->shap) (1.16.0)
Using cached shap-0.50.0-cp311-cp311-win_amd64.whl (547 kB)
Installing collected packages: shap
Successfully installed shap-0.50.0
```

[1]:
```python
import shap
shap.__version__
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[1], line 1
----> 1 import shap
      2 shap.__version__

File ~\anaconda3\Lib\site-packages\shap\__init__.py:1
----> 1 from ._explanation import Cohorts, Explanation
      3 # explainers
      4 from .explainers import other

File ~\anaconda3\Lib\site-packages\shap\_explanation.py:10
```

```
      7 from typing import TYPE_CHECKING, Any, cast
      9 import numpy as np
---> 10 import pandas as pd
     11 import scipy.cluster
     12 import scipy.sparse

File ~\anaconda3\Lib\site-packages\pandas\__init__.py:22
     19 del _hard_dependencies, _dependency, _missing_dependencies
     21 # numpy compat
---> 22 from pandas.compat import is_numpy_dev as _is_numpy_dev  # pyright:␣
 ↪ignore # noqa:F401
     24 try:
     25     from pandas._libs import hashtable as _hashtable, lib as _lib, tsli␣
 ↪as _tslib

File ~\anaconda3\Lib\site-packages\pandas\compat\__init__.py:18
     15 from typing import TYPE_CHECKING
     17 from pandas._typing import F
---> 18 from pandas.compat.numpy import (
     19     is_numpy_dev,
     20     np_version_under1p21,
     21 )
     22 from pandas.compat.pyarrow import (
     23     pa_version_under1p01,
     24     pa_version_under2p0,
   (…)
     31     pa_version_under9p0,
     32 )
     34 if TYPE_CHECKING:

File ~\anaconda3\Lib\site-packages\pandas\compat\numpy\__init__.py:4
      1 """ support numpy compatibility across versions """
      2 import numpy as np
----> 4 from pandas.util.version import Version
      6 # numpy versioning
      7 _np_version = np.__version__

File ~\anaconda3\Lib\site-packages\pandas\util\__init__.py:2
      1 # pyright: reportUnusedImport = false
----> 2 from pandas.util._decorators import (  # noqa:F401
      3     Appender,
      4     Substitution,
      5     cache_readonly,
      6 )
      8 from pandas.core.util.hashing import (  # noqa:F401
      9     hash_array,
     10     hash_pandas_object,
     11 )
```

```
     14 def __getattr__(name):

File ~\anaconda3\Lib\site-packages\pandas\util\_decorators.py:14
      6 from typing import (
      7     Any,
      8     Callable,
      9     Mapping,
     10     cast,
     11 )
     12 import warnings
---> 14 from pandas._libs.properties import cache_readonly
     15 from pandas._typing import (
     16     F,
     17     T,
     18 )
     19 from pandas.util._exceptions import find_stack_level

File ~\anaconda3\Lib\site-packages\pandas\_libs\__init__.py:13
      1 __all__ = [
      2     "NaT",
      3     "NaTType",
   (…)
      9     "Interval",
     10 ]
---> 13 from pandas._libs.interval import Interval
     14 from pandas._libs.tslibs import (
     15     NaT,
     16     NaTType,
   (…)
     21     iNaT,
     22 )

File ~\anaconda3\Lib\site-packages\pandas\_libs\interval.pyx:1, in init pandas.
 ↪_libs.interval()

ValueError: numpy.dtype size changed, may indicate binary incompatibility.␣
 ↪Expected 96 from C header, got 88 from PyObject
```

[ ]:

```
```